

Lab 06: OpenMP Parallel Programming

Exercise 04 Report: Loop Parallelization & collapse(2)

1. Exercise Goal

Evaluate standard loop parallelization versus collapse(2).

2. Source Code

```
#include <omp.h>
#include <stdio.h>

int main() {
    printf("=== Collapsed #pragma omp parallel for collapse(2) ===\n");
    #pragma omp parallel for collapse(2)
    for (int r = 1; r <= 5; r++) {
        for (int k = 1; k <= 10; k++) {
            printf("r = %d, k = %d, thread # = %d\n", r, k, omp_get_thread_num());
        }
    }
    return 0;
}
```

3. Explanation & Analysis

Without collapse, OpenMP only parallelizes the outermost loop ($r = 1..5$). The 5 iterations of loop r are split among the available threads.

The collapse(2) clause combines outer loop r (5) and inner loop k (10) into a single 2D iteration space of 50 iterations, which are distributed evenly across threads for better load balancing.

5. Execution Output Screenshot

